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A paleomagnetic investigation of the Lower Ordovician St. George Group, Port-au-Port Peninsula, Newfoundland: implications for the Iapetus Ocean and evidence for Late Paleozoic remagnetization

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ABSTRACT

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To obtain an early Ordovician paleomagnetic pole position for North America, and to determine the paleolatitude of its margin of the Iapetus Ocean prior to the Taconic orogeny, a study of the lower St. George Group (Tremadocian–Arenigian) of western Newfoundland was undertaken. Analysis of 84 oriented core samples from six localities on Port au Port Peninsula indicates that the NRM of the St. George rocks is the resultant of three distinct paleomagnetic directions. The first is a steep downward component commonly observed in the low temperature range of demagnetization (room temperature 250 °C) in virtually all samples, that roughly corresponds to the present day axial dipole field direction *in situ* ($D = 350°$, $I = 68°$). The second is a stable component with southerly declinations and shallow-intermediate negative inclinations (D, $I_{\text{in situ}} = 176°$, $-20°$; D, $I_{\text{tilt corrected}} = 173°$, $-5°$, respectively) usually seen in the range of 450–600 °C and found in approximately 46% of the specimens. The third is a stable southeasterly, shallow-intermediate positive component (D, $I_{\text{in situ}} = 161°$, $16°$; D, $I_{\text{tilt corrected}} = 163°$, $34°$, respectively) generally unblocked within the 350–550 °C or 450–600 °C range which was observed in 40% of the specimens. These components of remanence are herein referred to as the Type 1, Type 2, and Type 3 components, respectively. The remaining 14% of the specimens were found to be either directionally unstable to thermal demagnetization or totally overprinted by the Type 1 component. The Type 2 component is interpreted to represent a post-folding, Late Paleozoic remagnetization possibly associated with the Alleghanian Orogeny. On the other hand, the Type 3 component is interpreted as an Early Ordovician, possibly pre-Taconic remanence and, in tilt corrected coordinates, yields a paleolatitude for this region of the North American craton of 19 °S ± 6 °.

Introduction

Most models of the early development of the Appalachian Orogen in Newfoundland require the formation and ultimate destruction of a Precambrian to Early Paleozoic ocean: the Iapetus ocean (Dewey, 1969; Wilson, 1966; Stevens, 1970; Williams, 1978, 1980). The width of this ocean, prior to its closure, is a matter of some debate, but estimates based on paleomagnetic evidence (Johnson and Van der Voo, 1985; Briden et al., 1988), and palinspastic restorations (Williams, 1980) yield a minimum value between 1000 and 2000 km. Faunal evidence indicates an ocean width of as much as 3500 km (Neuman, 1984). Furthermore, paleomagnetic evidence suggests that the northern (N. American) margin of this ocean had an approximately E–W orientation and was located in equatorial latitudes during most of the Early Paleozoic (Van der Voo, 1988).

The Lower Ordovician (Tremadocian–Arenigian) St. George Group of the Port-au-Port, Peninsula of western Newfoundland (Fig. 1) is interpreted as a remnant of extensive Early

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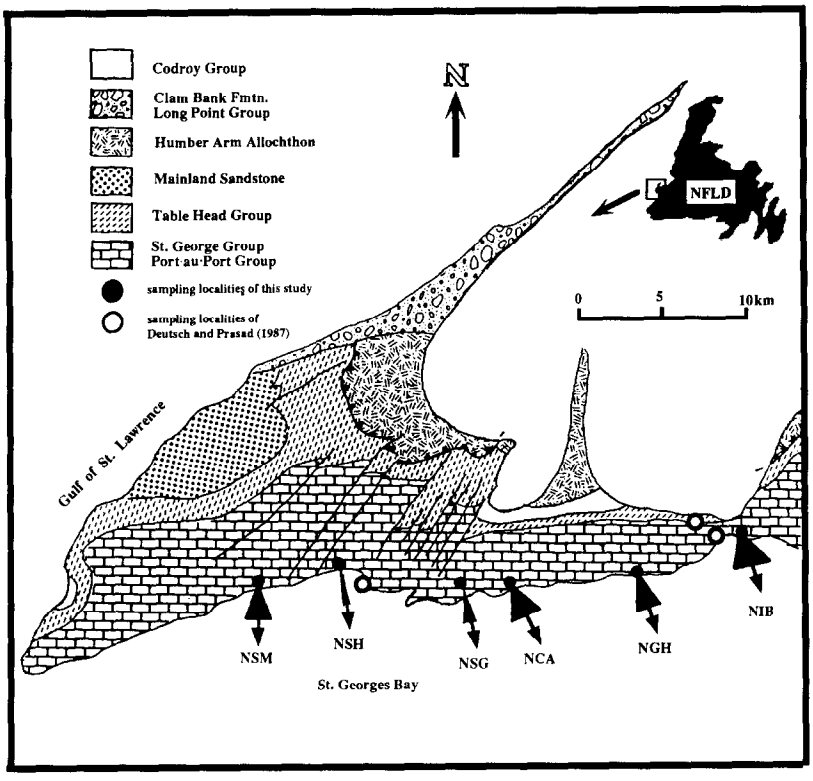


Fig. 1. Generalized geologic map of Port-au-Port Peninsula, western Newfoundland (after James and Stevens, 1982) showing the six sampling sites from which the samples used in this study were collected. Arrows at each locality indicate the direction of the mean declination of the Type 3 component after structural correction (see text for discussion). Blackened areas are the approximate amount of error associated with each determinations. Also shown are the sampling sites of Deutsch and Prasad (1987).

Paleozoic shallow-water carbonate sequences that were deposited along the continental margin of North America facing the Iapetus Ocean (Pratt and James, 1986). These rocks suffered deformation during the Taconic, Acadian, and to a lesser degree, the Alleghanian orogenies and now form an integral part of the autochthonous and parautochthonous stratigraphy of the northern Appalachians (Williams, 1978).

This investigation was undertaken primarily to determine the paleolatitude of the margin prior to the closure of the Iapetus, and to document the timing of deformation and the amount of any tectonic rotations which may have occurred during the deformation. Previous paleomagnetic results from Ordovician rocks of western Newfoundland include those of Beales et al. (1974), Deutsch and Rao (1977), Deutsch and Prasad (1987), and Hall and Evans (1988). The Hall and Evans (1988) study presented preliminary results from the St. George Group of Port-au-Port and these data are included in this study. Deutsch and Prasad (1987) also analyzed specimens of the St. George Group from Port-au-Port. Their sampling, although intensive, was restricted to two localities. Here, we report results from samples collected from the lower St. George Group at six localities along the southern coast of Port-au-Port Peninsula which complement and extend the existing Ordovician paleomagnetic data base for western Newfoundland.

Tectonic setting

Three main tectonic/orogenic events are observed in the geologic record of the northern Appalachians; the Taconic Orogeny (Middle–Late Ordovician), the Acadian Orogeny (Devonian), and the Alleghanian Orogeny (Carboniferous–Permian). The timing of these deformational episodes is directly related to the destruction of the

Iapetus Ocean, deformation of its margins, and continental accretion through the suturing and incorporation of more outboard terranes into the orogen (Williams and Hatcher, 1983).

The Appalachian Orogen of the Canadian Maritimes can be divided into five general tectono-stratigraphic zones (Williams, 1978; Williams, 1979, 1980; Williams and Hatcher, 1983). From west to east these are the (1) Humber Zone, (2) Dunnage Zone, (3) Gander Zone, (4) Avalon Zone, and (5) Meguma Zone. The Humber Zone of western Newfoundland contains a record of the development of a stable "Atlantic type" continental margin of the Iapetus Ocean and its subsequent deformation during the Taconic Orogeny due to the collision of the Laurentian margin with an oceanic island arc terrane (the Dunnage Zone) during the Early Paleozoic (Williams, 1980; Casey and Dewey, 1984; Casey et al., 1985). This pre-terminal collision resulted in the foundering of the previously stable continental shelf and obduction of the Humber Arm Allochthon onto the margin of the craton (Rodgers and Neale, 1963; Rodgers, 1970; Williams, 1975, 1979). The amount of paleolatitudinal separation that existed between the Humber Zone and the island arc terrane is unclear, but given the disparity between the age of Iapetus rifting (600 Ma) and the age of ophiolites within the allochthon (508 Ma), the intervening oceanic tract would have been quite extensive even with slow rates of spreading (Casey and Dewey, 1984; Casey et al., 1985).

The autochthonous part of the Humber Zone is characterized by Cambro-Ordovician rocks thought to have been deposited as trailing margin sequences in sub-tropical epeiric sea conditions. The St. George Group (Schuchert and Dunbar, 1934; Knight and James, 1987) of the Humber Zone in western Newfoundland is a carbonate sequence that is entirely of Canadian age (Tremadocian–Arenigian) and forms part of the "outer limestone facies" of North American Lower Ordovician shallow-water carbonates (Knight and James, 1987). Equivalents of the Cambro-Ordovician carbonate sequences observed in the Humber Zone of western Newfoundland are preserved in the U.S. Appalachians, northwestern Scotland, and eastern Greenland (Williams, 1984).

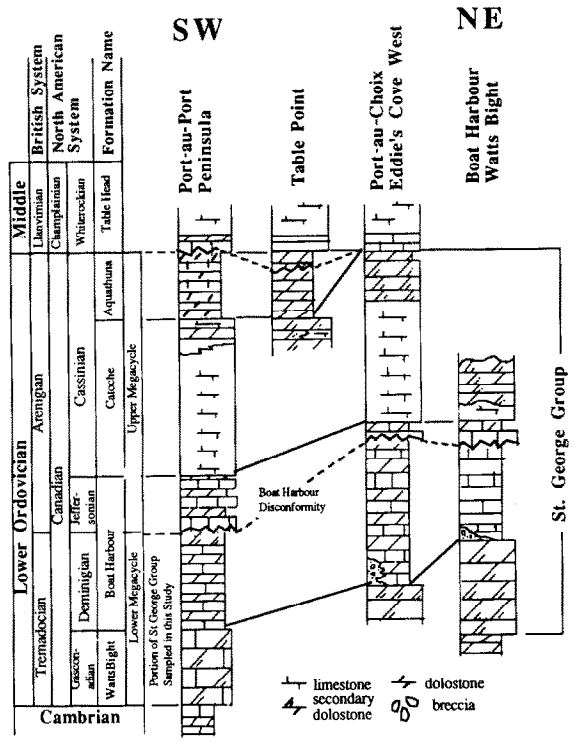


Fig. 2. Schematic stratigraphic subdivisions of the St. George Group in western Newfoundland (after Knight and James, 1987).

Geology and stratigraphy of the lower St. George Group, Port-au-Port Peninsula

On Port-au-Port Peninsula (Fig. 1) over 500 m of the St. George Group outcrops in coastal exposures and sea cliffs (James and Stevens, 1982; Pratt and James, 1982; Knight and James, 1987). The base of the St. George Group is marked by lower Tremadocian, bioturbated, dark colored limestones, diagenetic dolostones and associated cryptalgal boundstones which overlie lighter colored limestones and syngenetic dolostone of Late Cambrian age. The top of the section is marked by the Table Head–St. George disconformity, a horizon at which bedded limestones of the Table Head Group truncate cyclic dolostones, limestones, and shales of the upper St. George strata (Knight and James, 1987).

Figure 2 shows the most recently revised stratigraphic divisions of the St. George Group (Knight and James, 1987), and compares the section measured on Port-au-Port peninsula with those from

more northern regions of the Humber Zone. This revised interpretation (see James and Stevens, 1982; Pratt and James, 1982) subdivides the Lower Ordovician strata into four individual stratigraphic units. In ascending order these are the Gasconadian Watts Bight Formation, the Demingian and Jeffersonian Boat Harbour Formation, the Lower–Middle Cassinian Catoche Formation, and the Upper Cassinian Aguathuna Formation.

The Watt's Bight Formation contains the Green Head bioherm (NGH in Fig. 1) described in James and Stevens (1982), and Pratt and James (1982). The Green Head section is a 12 m thick, highly complex amalgamation of algal-metazoan mounds some of which are several meters in diameter, and contain Lichenaria corals, and Renalcis (moss-like, calcified Algae), and cephlapods of Gasconadian or earliest Tremadocian age (*Diphragmoceras* beds of Flower, 1978). This well dated stratigraghic interval was used as a reference section during paleomagnetic sampling of the St. George Group at other localities on Port-au-Port Peninsula (Fig. 1 shows the locations of five other sampling sites). The entire thickness of the St. George Group is contained in separate outcrop sections on the Port-Au-Port Peninsula with the lower sections predominantly exposed along the southern shore (Fig. 1). The strike of the bedding here roughly parallels the approximately E–W trend of the coastline. The affects of the Taconic Orogeny within this part of the authochthon are unclear, but the gentle northward regional tilt of the bedding observed is primarily attributed to the Acadian Orogeny (Devonian—Schillereff and Williams, 1979; Cawood and Williams, 1988). Northeast–southwest trending high-angle faults of Carboniferous? age dissect the entire autochthonous sequence as well as truncate the basal thrust of the Humber Arm Allochthon between Picadilly Head and Harry Brook (Schillereff and Williams, 1979). Karst deposits, other than those associated with the Boat Harbour disconformity, occur locally within the St. George Group as collapse breccias deposited within former sinkholes or caves. Near Isthmus Bay, these karst deposits extend down section from the Table Point Formation well into the lower St. George and have been classified as Carboniferous in age (Schillereff and Williams, 1979; Knight and James, 1987). These features, in addition to the high-angle faults within the autochthon suggest that this section was uplifted to shallow crustal levels during the Late Paleozoic Alleghanian Orogeny, and was further subjected to fairly extensive circulation of meteoric waters through subterranean conduits.

Paleomagnetic study

Sampling

Eighty-four oriented core samples were collected from six sampling sites along the southern coast of Port-au-Port Peninsula, western Newfoundland. The locations of these sites are shown in Fig. 1 and the number of samples collected from each (n_0) are listed in Table 1. All specimens were obtained using a gasoline powered coring drill and were oriented in geographic coordinates by standard methods utilizing a Brunton compass and leveling table. The largest sampling population was drawn from the stratigraphically well dated Green Head Complex (NGH in Fig. 1). The remainder of the sample collecting was stratigraphically controlled by using the NGH site as a "reference" section. The core length of most samples was sufficient to allow "sister" specimens to be cut from a single core. There is little structural variation between the six sites as the bedding uniformly dips gently ($\approx 20°$) to the north-northwest, precluding the application of a fold test.

Laboratory techniques

Standard paleomagnetic and rock magnetic experiments were carried out primarily at the Paleomagnetics Research Laboratory of the University of Houston, and also at the laboratory of the University of Texas at Austin. The objectives of the experiments were to determine the magnetic components comprising the remanence, and the mineralogical phases in which they reside. To achieve these objectives, samples from each site were analyzed by progressive thermal demagnetization of the NRM, standard IRM acquisition

TABLE 1

Site level mean directions of the Type 1, 2 and 3 remanence components *

Site	n/n_0	In situ		Tilt corrected		k	α_{95}
		D (°)	I (°)	D (°)	I (°)		
Type 1 site means							
NIB	10/10	342.0	58.9	n.l.	n.l.	18.9	11.4
NGH	23/27	353.1	68.1	n.l.	n.l.	8.5	10.3
NCA	6/10	296.6	72.9	n.l.	n.l.	17.9	16.2
NSG	9/9	351.6	66.5	n.l.	n.l.	52.07	7.2
NSH	12/12	354.7	63.9	n.l.	n.l.	52.07	7.2
NSM	0/16			n.l.	n.l.		
Type 2 site means							
NIB	3/10	175.6	−29.7	166.2	−10.0	6.07	55.0
NGH	6/27	186.4	−16.6	184.8	2.9	11.7	20.4
NCA	3/10	180.0	−17.1	179.0	−6.9	15.8	32.0
NSG	3/9	175.9	−9.0	176.7	−2.1	128.1	10.9
NSH	9/12	164.9	−20.2	164.7	−6.8	8.6	18.6
NSM	15/16	176.0	−21.7	173.2	−6.0	17.6	9.3
Type 3 site means							
NIB	7/10	156.0	5.6	159.3	30.7	8.9	21.3
NGH	12/27	161.7	20.9	161.9	44.2	12.5	12.7
NCA	5/10	156.1	20.3	156.2	31.6	11.5	23.5
NSG	7/9	163.3	19.4	167.4	29.6	35.1	10.3
NSH	1/12	168.4	−1.0	168.6	13.6		
NSM	2/16	176.6	0.0	178.6	19.0	399.2	12.5

* Location of sites are shown in Fig. 1. n/n_0 is the number os samples exhibiting a particular directional component/the number of samples demagnetized; k = the precision parameter; α_{95} = the radius of a cone of 95% confidence about the mean. n.l. = tilt corrected directions not listed.

and thermal demagnetization of IRM experiments, and techniques including constant field IRM acquisition combined with incremental heating. Alternating field demagnetization proved to be an ineffective cleaning method for these rocks, presumably due to the inclusion of high coercivity minerals in the magnetic grain fraction.

Magnetic measurements were made on a ScT cryogenic magnetometer housed in shielded room at both the University of Houston and the University of Texas at Austin. Measurement of test sets of specimens performed on each instrument provided repeatable results. Units of measurement for the cryogenic magnetometers were in emu/g with background noise levels on the order of 5×10^{-9} emu/g (W. Gose, pers. commun.) for those instruments. At the University of Houston, thermal demagnetization analysis was performed using a Schondstet Thermal Specimen Demagnetizer, model TSD-1 situated in a shielded laboratory. Thermal experiments done at the University of Texas at Austin were carried out with a similar instrument also located within a shielded room.

The isothermal remanence (IRM) experiments were of two types. The first followed standard procedures in which a specimen was exposed to incrementally higher magnetic fields at room temperature until either saturation magnetization, or the maximum available field level is reached. The standard IRM experiments were performed at the the University of Houston and utilized an electromagnet with a maximum output of 7 kG. Some of the IRM specimens were subsequently thermally demagnetized to determine the unblocking temperature (T_{UB}) spectra of the acquired IRM.

The second type of IRM experiment was designed to measure any changes in the coercivity spectra of the magnetic fraction of a particular

specimen as it was exposed to the repeated heating and cooling cycles of standard thermal demagnetization techniques. Keeping the applied field level constant, the procedure used was to allow the specimen to acquire an IRM following each progressively higher increment of thermal demagnetization. The purpose of this type of experiment was to qualitatively monitor the compositional changes within the magnetic fraction associated with thermal cleaning. The experiments involving alternating IRM acquisition and progressive thermal demagnetization steps were carried out at the University of Texas at Austin laboratory. An applied field of 5 kG was used for each IRM acquisition step while the thermal steps were as follows; 0°, 100°, 200°, 300°, 350°, 400°, 450°, 500°, 550°, 600°, 675° C.

Progressive thermal demagnetization of NRM

In general, thermal treatment of most of the St. George specimens provided J_{NRM} vs. T curves which, although quite variable, were indicative of thermally distributed T_{UB} spectra (Fig. 3). However, curves from most specimens showed evidence of the presence of a "soft" component in the low temperature range while the high temperature part of the T_{UB} spectra indicated that either magnetite, or hematite, or both form a major part of the natural remanence in these rocks. The range of demagnetization employed was sufficiently extensive as to either effectively reduce the NRM intensity of the samples or render them directionally unstable. For the suite of samples analyzed, thermal demagnetization revealed three distinct directional components of remanence. The temperature interval between successive demagnetization steps was such that at least two of these three remanence components could be resolved from almost all specimens which responded favorably to demagnetization. Finally, as Fig. 3 shows, there appears to be no strict correlation between the shape of the J_{NRM} vs. T curves and the type of remanence components exhibited by the representative specimens.

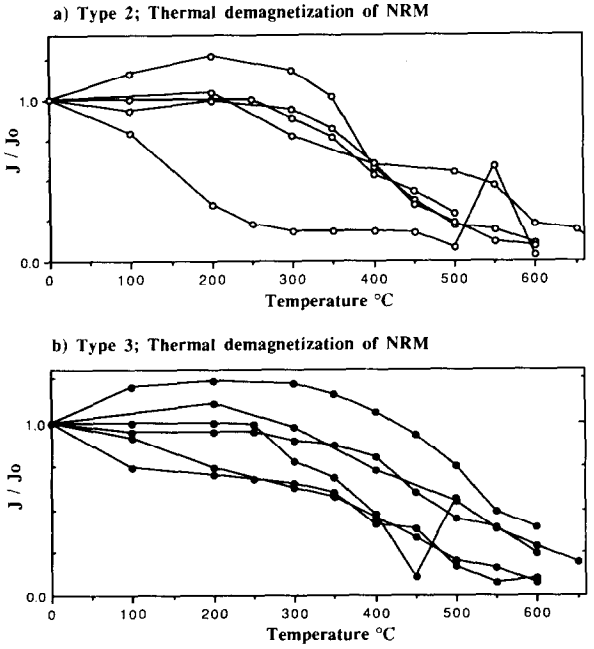


Fig. 3. Thermal demagnetization of NRM curves for 12 representative samples; (a) Type 2 samples, (b) Type 3 samples.

Directional analysis from thermal demagnetization

Generally speaking, the discrimination between the three remanence components at the specimen level was facilitated via visual inspection of demagnetization paths as displayed in orthogonal and stereographic projection in combination with principal components analyses of these demagnetization paths. Following Kirschvink (1980), the maximum angle of deviation (MAD) was calculated for each vector direction resolved from principal component analysis. For the most part, the discrimination was based upon the inclination values, and, to a lesser degree, declinations of each specimen. The Type 1 component is distinguished by extremely steep, downward inclinations and northerly declinations both in situ and after structural correction. Also, the presence of the Type 1 component is restricted to the low temperature range of thermal treatment. The Type 2 component characteristically exhibits southerly vector directions with negative (upward) inclinations in situ and either negative or nearly horizontal inclinations in structurally corrected coordinates. The Type 3 component is, in general, characterized by southeasterly declinations and positive (downward) inclinations before and after application of the structural correction. The slopes

of Type 3 demagnetization paths in situ indicate shallow positive inclinations which steepen to more intermediate angles in structurally corrected coordinates. Furthermore, the Type 2 component seems to differ in declination from the Type 3 in that the former tend to plot in a more SSW direction while the latter generally fall in the SE quadrant. No strict distinction between Type 2 and Type 3 components could be made solely on the basis of the temperature range in which they were observed due to partially overlapping T_{UB} spectra. Finally, while the Type 1 component was often observed in association with either the Type 2 or Type 3 component, the simultaneous occurrence of Type 2 and Type 3 components within an individual specimen was seldom found.

The handling of vector data at the formation level can be accomplished either by calculating the mean of specimens from all sites that yield linear components (where n = number of specimens) or by finding the mean of the site level means (where n = number of sites). For this suite of rocks, the first method was chosen for the following reasons: (a) there is little variability in the structural orientation of the bedding between sites, (b) all sites are located on the southern coast of Port-au-Port peninsula and are within a degree of one another in both latitude and longitude, (c) the number of samples from a given site exhibiting a particular type of remanence component is highly variable, and often quite small. Site mean vector data for Type 1, 2, and 3 components are listed in Table 1. Pole positions and associated Fisher statistics are shown in Table 2.

The NRM of virtually all samples exhibited evidence for contribution from the Type 1 component in the low to intermediate unblocking temperature range. Inspection of orthographic plots (Figs. 4 and 5) reveal that while this component is unblocked over a variable range of temperatures, it is consistently northerly directed and steeply positive *in situ*. This component was observed and resolved in a total of 60 specimens. Mean Type 1 vector directions from each site are shown in Table 1. In geographic coordinates this component corresponds fairly well to the present day field and probably represents a recent magnetization. For this reason, the tilt corrected data are not listed in Table 1. Figure 6 shows the stereographic projection of the population of vector directions from which the Type 1 mean was drawn. The mean vector direction was calculated to be $D = 350°$, $I = 68°$ and the corresponding Fisher statistics for $n = 60$ are $R = 55.93$, $k = 14.5$, and $\alpha_{95} = 5°$.

Figure 4 shows orthogonal vector plots for samples from each site that exhibited demagnetization paths characteristic of the Type 2 component. These plots show that following the removal of a low T_{UB} component, the decay paths are marked by negative, southerly directions in the intermediate to high temperature range. The Type 2 component was most often observed above 450 °C, but was also revealed over a broader temperature range in a few cases (Fig. 4e, f). Figure 4a shows the behavior of specimen NIB 2 which proved to be one of the few examples for which resolution of all three remanence components coexisting in the

TABLE 2

Paleomagnetic pole positions and Fisher statistics for the Type 1, 2, 3 remanence components *

Remanence component	In situ							Tilt corrected					
	N	R	k	α_{95} (°)	pole	D_p (°)	D_m (°)	R	k	α_{95} (°)	pole	D_p° (°)	D_m (°)
Type 1	60	55.93	14.5	4.9	83° N, 130° W	6.9	8.3	n.l.	n.l.	n.l.	n.l.	n.l.	n.l.
Type 2	39	36.04	12.9	6.6	52° S, 52° W	3.6	7.0	35.93	12.4	6.8	44° S, 50° W	3.4	6.8
Type 3	34	31.40	12.7	7.2	31° S, 37° W	3.8	7.4	31.46	13.0	7.1	21° S, 42° W	4.6	8.1

* N = total number of samples used to calculate the pole; R = the length of the resultant vector; k = the precision parameter; α_{95} = the radius of a cone of 95% confidence about the mean; D_p, D_m = the semiaxes of an oval of 95% confidence about the poles. n.l. = tilt corrected Type 1 pole not listed (see text).

natural remanence of a single specimen appeared to be possible. Though not well constrained, principal component analysis of the components contained in NIB 2 suggests the presence of three linear trends of differing orientations between 0 °C and 200 °C (MAD = 3.1°), between 300 °C and 400 °C (MAD = 1.9°), and between 500 °C and 600 °C (MAD = 1.8°). The steep downward, and northerly directed component observed in the low temperature range corresponds to a Type 1 overprint ($D = 360°$, $I = +54°$; in situ). The south-southwesterly, and shallow, negative trend in the intermediate temperature range (300–450 °C) represents the Type 2 component ($D = 187°$, $I = -15°$; in situ and $D = 2°$, tilt corrected). The high temperature portion of the demagnetization path forms a linear trend toward the origin of the vector plot and appears to correspond to the Type

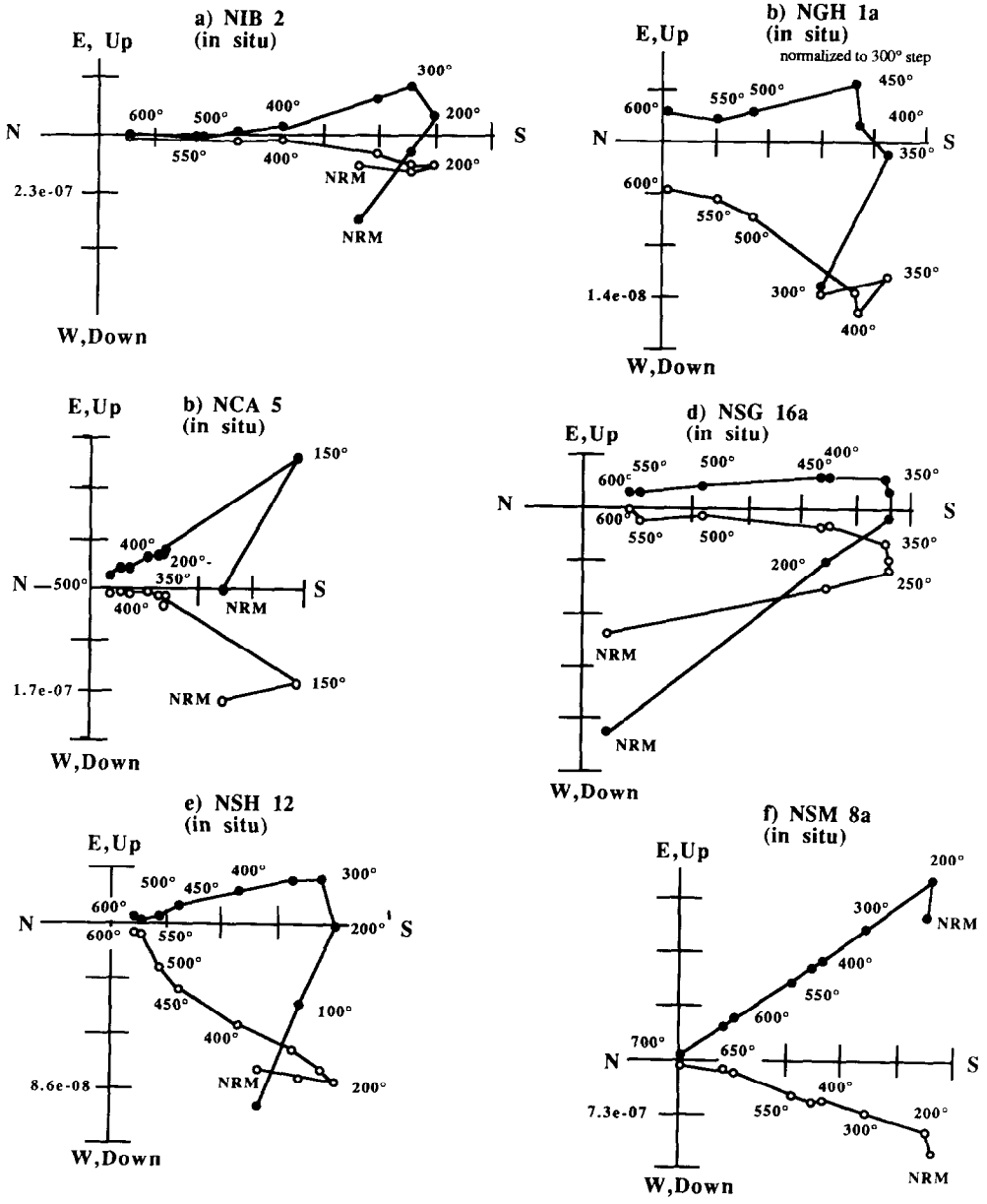


Fig. 4. Orthogonal projections for six representative samples exhibiting Type 2 remanence components. Circles = horizontal projection, dots = vertical projection. Intensities are in emu/g.

3 component ($D = 185°$, $I = 3°$, in situ, and $D = 189°$, $I = +19°$; tilt corrected).

The in situ and structurally corrected site mean vector data for the Type 2 component determined from the six sites are listed in Table 1. Figure 7 displays the stereographic projection of characteristic directions determined from 39 specimens exhibiting this component. Although there exists a considerable amount of "stringing" in the SE and SW quadrants, the distribution has a density maximum slightly east of due south and near the perimeter on the upper hemisphere of the stereonet. *In situ*, all points plot on the upper hemisphere and the maximum is located in a more steeply negative position than that of the structurally corrected data. The in situ and structurally corrected mean vector directions drawn from this population ($n = 39$) are found to be $D = 176°$, $I = -20°$ ($R = 36.04$, $k = 12.9$, and $\alpha_{95} = 6.6°$) and $D = 173°$, $I = -5°$ ($R = 35.93$, $k = 12.4$, and $\alpha_{95} = 6.8°$), respectively. The Fisher statistics associated with the mean determination are relatively unchanged by the bedding correction.

Figure 5 shows orthogonal vector diagrams for specimens from each site which exhibited Type 3 characteristic directions. These plots indicate that the "soft" component is removed at temperatures below 350° in favor of a more stable, southeasterly and positive (downward) component corresponding to the Type 3 direction. Generally, this remanent component was seen to have a T_{UB} spectrum similar to that of Type 2 and persisted

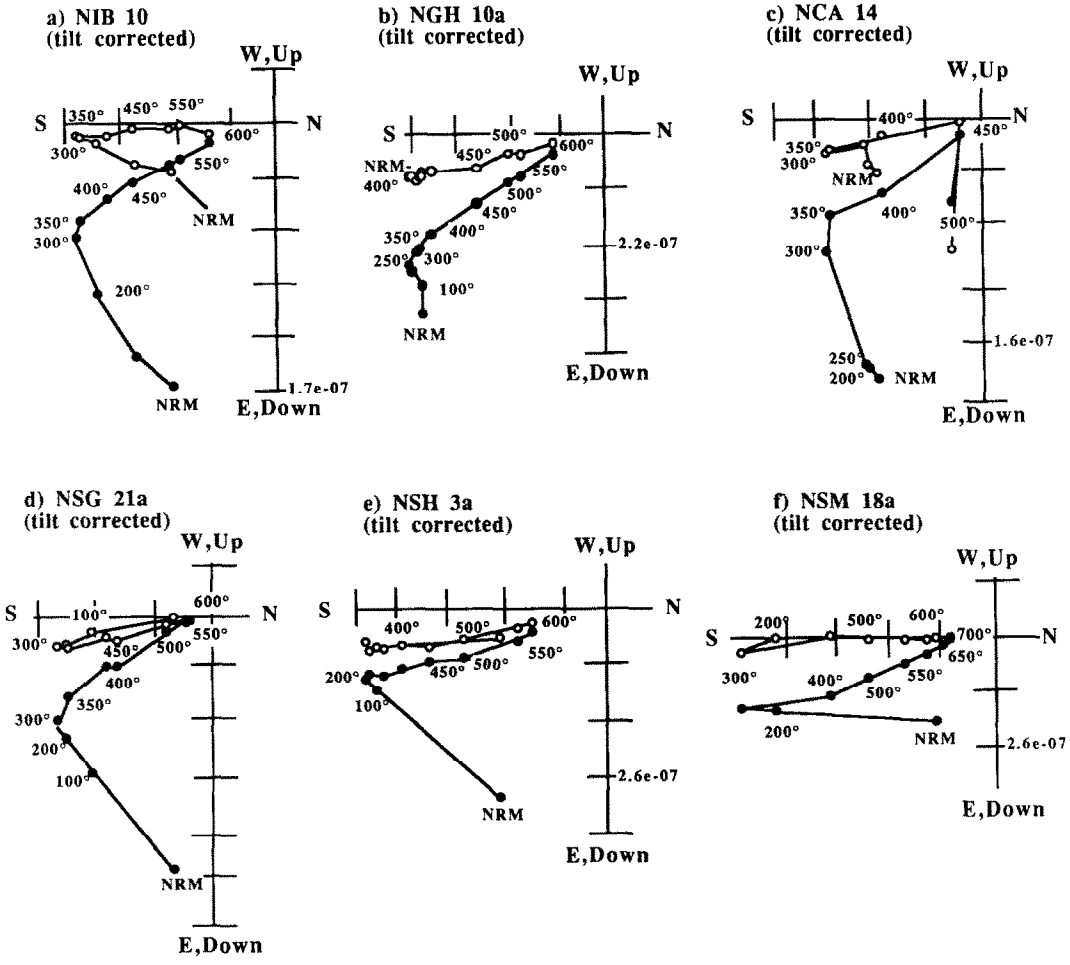


Fig. 5. Orthogonal projections for six representative samples exhibiting Type 3 remanence components. Circles = horizontal projection, dots = vertical projection. Intensities are in emu/g.

to 550° or 600°C. However, as Fig. 5c illustrates, there were exceptions to this generalization. The demagnetization path for this specimen (NCA 14) revealed the Type 3 component between 350° and 450°C and indicates that the natural remanence is randomized at the 500°C step.

Figure 7 also displays the characteristic directions determined from 34 specimens exhibiting Type 3 remanence. With the exception of two anomalous points, the data fall in a "swath" of points in the SE quadrant and despite the degree of dispersion, determination of the resultant (R) was found to be significant at the 95% level. The mean vector direction ($n = 34$) was calculated to be $D = 161°$, $I = 16°$ ($R = 31.40$, $k = 12.7$, $\alpha_{95} = 7.2°$) without tilt correction; or $D = 163°$, $I = 34°$ ($R = 31.46$, $k = 13.0$, and $\alpha_{95} = 7.1°$) with tilt correction. As can be seen, the declination is little changed by the structural correction applied. This is because its orientation is aligned sub-parallel to the direction of the strike of bedding and rotation of bedding planes to horizontally only steepens the inclination. Additionally, the Fisher statistics associated with the determination before and after correction are approximately equal.

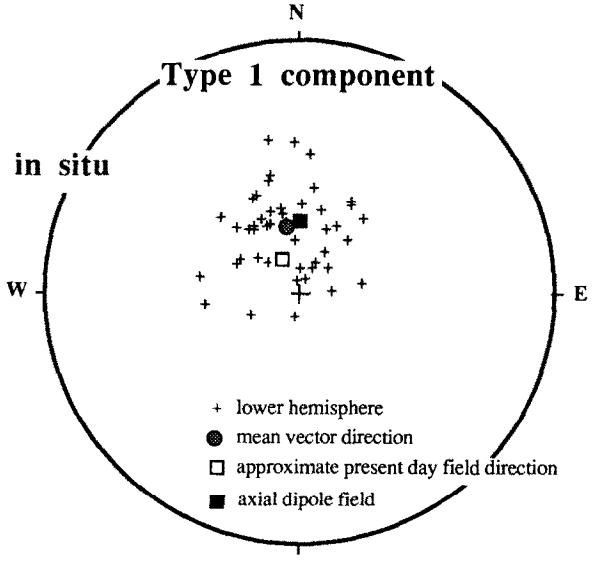


Fig. 6. Stereographic projection of Type 1 remanence directions *in situ*. Also shown are the locations of the mean direction, the approximate present day field at the site, and the axial dipole field.

IRM experiments

Wafer sized specimens from core samples of each site were selected for standard IRM acquisition, and thermal demagnetization of IRM experiments on the basis of the type of behavior exhibited by the sample during NRM analysis. These were chosen to be representative of Type 2 and Type 3 samples from a particular site. An additional, but smaller subset of the representative specimens were chosen for the alternating IRM acquisition and incremental heating experiments.

The objective of the standard IRM acquisition experiments was to analyze the bulk coercivity properties of the NRM assemblage of magnetic phases contained within representative specimens. In this way, evidence for the mineralogical composition of the assemblages most likely to be responsible for the type of remanence exhibited could be obtained. Thermal demagnetization of the IRM imparted to these specimens was carried out to determine the T_{UB} spectra of minerals important in the acquisition of the induced magnetization. This type of rock magnetic analysis was extremely important given the similar T_{UB} ranges observed for Type 2 and Type 3 components, and the non-discrete shape of the J_{NRM} vs. T curves for most samples.

A total of 22 specimens were used for the acquisition experiments including 12 which were representative of the Type 2 component and 10 which were representative of the Type 3 component. The results show that of the 12 Type 2 specimens, 10 exhibited evidence for the predominance of high coercivity minerals, 2 exhibited evidence of the presence of both high and low coercivity minerals. None of the Type 2 specimens displayed IRM behavior indicative of the presence of low coercivity minerals alone. Of the 10 Type 3 representatives analyzed, 6 specimens indicated high coercivity properties only, 3 specimens exhibited strictly low coercivity behavior , and 1 specimen exhibited evidence for both high and low coercivity mineralogical phases. Figure 8a shows examples of the IRM acquisition curves and illustrates the observed acquisition behavior.

The subsequent thermal demagnetization of the IRM was carried out on 18 specimens including 9

representatives of both Type 2 and Type 3 specimens. Each Type 2 demagnetization curve indicates that hematite was the mineral most important in the acquisition of the IRM induced in these specimens. In Figure 8b, the two specimens exhibiting low saturation potential (open symbols) indicate that a significant portion of the IRM is contained in minerals with T_{UB}'s in excess of 575°C. Of the Type 3 specimens, three curves clearly display the reduction of the J_{IRM} to zero with the unblocking of magnetite. This is displayed in Fig. 8b, in which the three specimens exhibiting low bulk coercivity properties in Fig. 8a are completely demagnetized between the 550° and 575°C steps. The remaining 6 Type 3 curves provide evidence for the presence of hematite. In Fig. 8a, the specimen yielding an acquisition curve indicative of the presence of both high-and low-

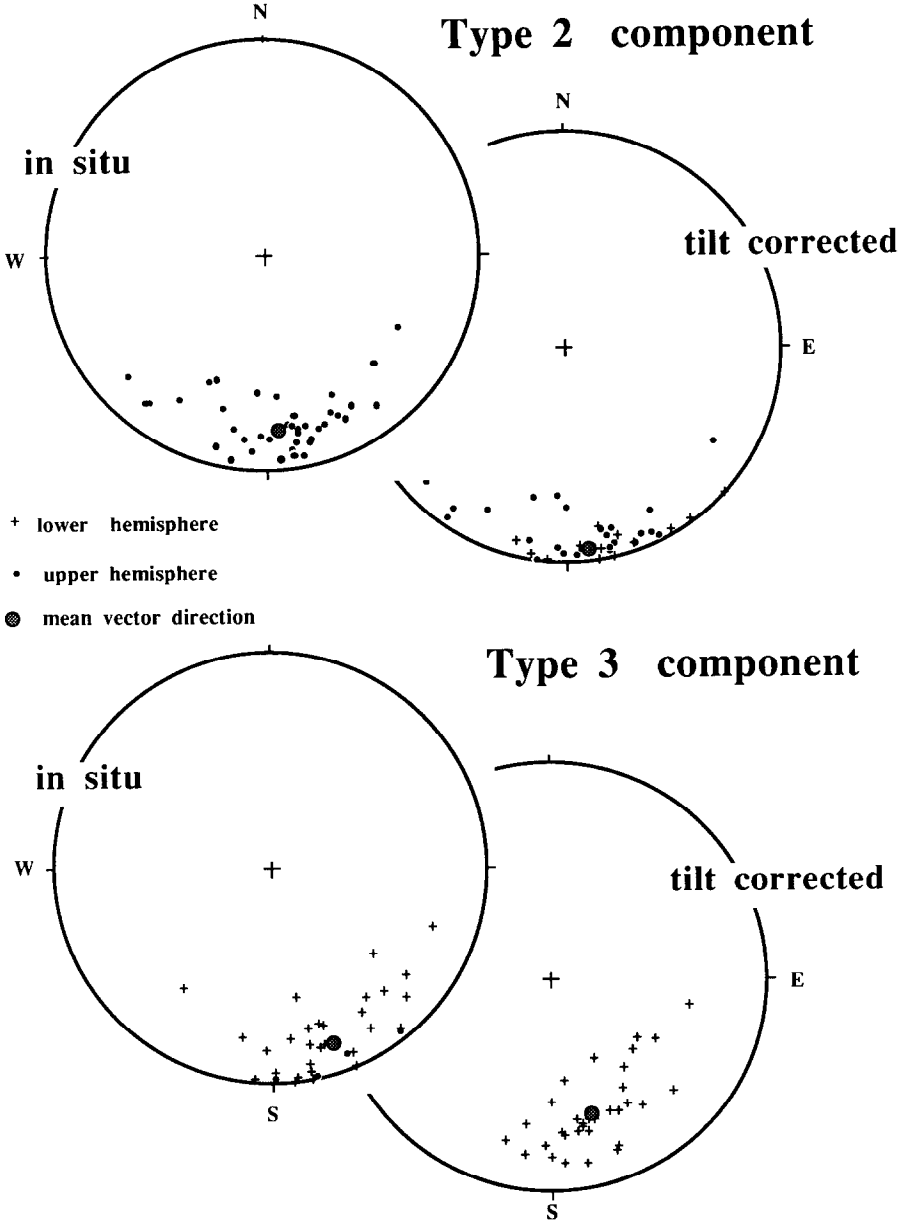


Fig. 7. Stereographic projection of Type 2 and 3 remanence directions before and after tilt correction. Also shown are the locations of the mean directions for each.

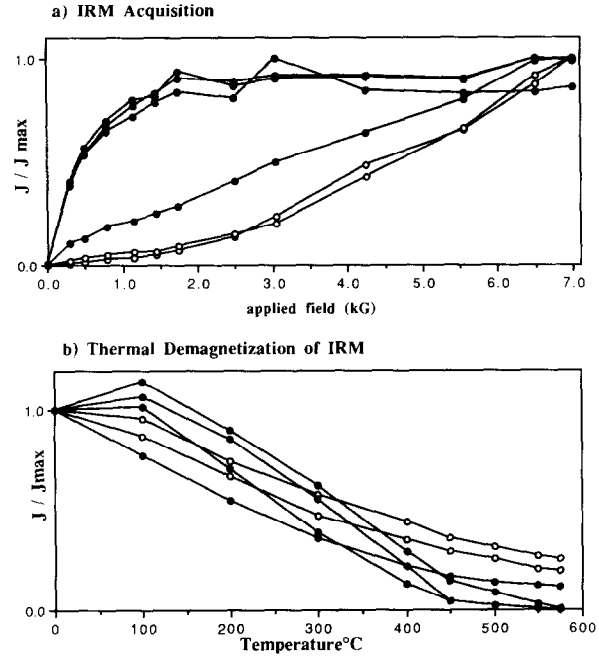


Fig. 8. Intensity of IRM plots for six representative specimens; (a) IRM acquisition curves, (b) thermal demagnetization of IRM curves. Open symbols = Type 2, solid symbols = Type 3.

coercivity phases, is characterized by a thermal demagnetization curve intermediate to the remaining five. Results from the thermal demagnetization are seen to correspond very well with those from the acquisition experiments and together, these indicate the following: (1) the Type 2 component is always associated with the presence of the high-coercivity mineral hematite, and (2) the Type 3 component is also found in association with hematite, and to a lesser degree, magnetite.

The purpose of the alternating IRM acquisition and incremental heating analysis was to monitor the changes in the magnetic properties which occurred in association with chemical reactions induced by the thermal demagnetization of the specimens. A total of 10 specimens were used for this analysis including 6 representative of Type 2 and 4 representative of Type 3. These methods aided in the detection of minerals both in the NRM assemblage and those induced thermally in particular representative specimens. However, there appears to be no real correlation between the IRM behavior as a function of increasing temperature and the type of remanence components exhibited by the specimens.

One characteristic common to all of the acquisition curves is a large increase in acquisition potential following heating to temperatures greater than 450° or 500°C reflecting an overall lowering of the bulk coercivity of the specimens at high temperatures. For 8 specimens, the acquisition curves reflect that, compared to the high temperature portion of the curve, bulk coercivity properties are relatively unchanged up to 450° or 500°C. The other two specimens yield acquisition curves which also exhibited an increase in J_{IRM} in the low-temperature range suggesting a significant contribution from goethite to the IRM properties. A common mineralogical alteration path inferred from the data includes the dehydration of goethite to form hematite at intermediate temperatures and the formation of low-coercivity minerals, such as magnetite, at the high temperature levels. One specimen from the NCA site (NCA 14; see Fig. 5c) showed behavior indicative of the formation of magnetite from pyrrhotite at intermediate temperatures.

Discussion of results

Comparisons between the results of this study and those from the earliest paleomagnetic investigations of the St. George Group (Beales et al., 1974; Deutsch and Rao, 1977) are difficult. In these early works, only A.F. demagnetization methods were used and may not have been adequate to completely isolate the stable remanence. These studies did not reveal the presence of more than one component, or report the techniques used to determine the sample directions. The Beales et al. (1974) study reported normal polarity directions ($D = 357°$, $I = -25°$) derived from A.F. demagnetization of St. George rocks collected near Daniel's Harbour, west Newfoundland (approximately 130 miles north of Port-au-Port Peninsula). However, the results of this study are problematical for several reasons. First, from their description of the rocks, it appears that their samples may have been collected from the upper part of the St. George Group. As such, it may not be appropriate to compare them directly to those derived for the Tremadocian part. Secondly, subsequent studies of the St. George Group (Deutsch

and Rao, 1977; Deutsch and Prasad, 1987; this study) have yielded only directions of reverse polarity, and results from the overlying Table Head Formation also indicate reversed directions (Deutsch and Prasad, 1987; Hall and Evans, 1988). The results of the Beales et al. study are anomalous in this regard, but it is possible that the remanence exhibited by the rocks at Daniel's Harbour records a brief change in the polarity of the Earth's geomagnetic field prior to the time of remanence acquisition. Third, it is unclear whether the directions reported were in in situ or structurally corrected coordinates. Knight and James (1987) indicate that rocks in this vicinity dip from 14° to 28° SW, but no mention of a tilt correction was made in the Beales et al. study. Finally, the samples were collected from a portion of the St. George Group strongly affected by Zn–Pb ore mineralization. Their study included an investigation of both the ore and the host rock (*pseudobreccia* of Cumming, 1968) and showed the paleomagnetic directions to be the same for both. The mineralization within this portion of the St. George Group has been attributed to the development of the Table Head–St. George disconformity (Cumming, 1968; Collins and Smith, 1975). Therefore, it is possible that the acquisition of the remanence exhibited by the St. George rocks at Daniel's Harbour occurred substantially later than the age of deposition, during a time of normal polarity. The location of the resulting anti-pole position (26°N, 126°E) within the Middle–Late Ordovician "hook" of the North American APW path of Van der Voo (1988; shown in Fig. 9) supports this speculation.

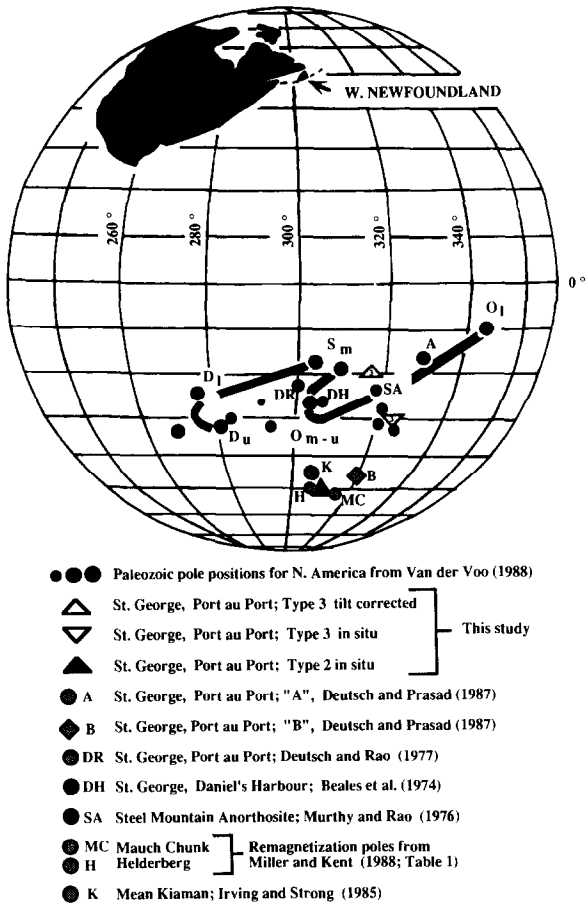


Fig. 9. APW path for North America during the Paleozoic after Van der Voo (1988) showing the positions of paleomagnetic poles determined in this study compared to those from other studies of Paleozoic rocks in western Newfoundland and elsewhere in the Appalachians. References to the APW path shown are in Table 1 of Van der Voo (1988). O_l = lower Ordovician, O_m = Middle Ordovician, O_u = Upper Ordovician, S_m = Middle Silurian, D_l = Lower Devonian, D_u = Upper Devonian.

Deutsch and Rao (1977) performed the first paleomagnetic investigation of the st. George Group on Port-au-Port Peninsula. The locations of the sites from which the samples were collected were not reported. In their study, Deutsch and Rao (1977) found stable directions from A.F. demagnetization that were nearly antipodal to those of Beales et al. (1974); ($D = 182°$, $I = 23°$; in situ). However, the authors state that A.F. demagnetization was ineffective in rocks containing hematite, that thermal methods were inconclusive, and found suitable demagnetization behavior in only 15 of 62 samples. As such, they reported direction only from rocks in which the remanence appeared to be contained in magnetite. The results of the present study indicate hematite to be nearly ubiquitous in these rocks on Port-au-Port Peninsula, and that magnetite is present only in samples that exhibit the Type 3 component. However, the Type 3 component differs substantially from the result determined by Deutsch and Rao. The significance of this difference is unclear, but may reflect contributions from undetected high coercivity phases in the NRM assemblage in their samples and the incomplete resolution of characteristic directions. The nearly antipodal relationship of

the Deutsch and Rao result to the Beales et al. (1974) result casts suspicions on the latter. Yet, as seen in Fig. 9, the Deutsch and Rao (1977) St. George pole lies outside of, but close to the Middle–Late Ordovician "hook" of the APW curve. Therefore, it can not be ruled out that both the Deutsch and Rao and Beales et al. results may reflect Middle Ordovician magnetizations not observed in the present study.

Murthy and Rao (1976) derived stable components of reversed directions ($D = 163°$, $I = 32.5°$; in situ) for the Steel Mountain anorthosites of the Grenvillian Province of southwestern Newfoundland (about 12 miles east of Port-au-Port Peninsula). Murthy and Rao attributed the origin of the observed remanence to a 451 ± 35 Ma thermal overprinting event inferred from K–Ar dating of these rocks. If correct, the age of remanence acquisition for these anorthosites would be Caradocian (early Late Ordovician). It should be noted that to assert that the in situ Steel Mountain direction represents a Caradocian magnetization is to assume these rocks have not suffered deformation since remanence acquisition. Given the pervasive effects of the Acadian orogeny (Williams and Hatcher, 1983; Cawood and Williams, 1988), this seems unlikely. Therefore, lack of structural control makes interpretation the Steel Mountain pole ambiguous. However, the Steel Mountain and tilt corrected Type 3 poles plot very near each other and both are close to the Early Ordovician part of the APW path in Fig. 9.

The paleomagnetic results from the St. George samples analyzed in the present study are found to be in general agreement with those of a more recent study of 85 specimens collected on Port-au-Port peninsula by Deutsch and Prasad (1987). While there is some overlap of the present work with the Deutsch and Prasad study, the sample suited used here was collected from a broader region along southern Port-au-Port in order to assess along strike variability. Figure 1 of Deutsch and Prasad (1987) shows that sampling in their study was concentrated at one locality near Isthmus Bay between the NGH and NIB sites and another to the west between the NSG and NSH sites (see Fig. 1). Those authors likewise discovered three components of remanence, a steep downward "soft" component and two reversed components of differing orientation designated the B, and A components. These are found to generally correspond to the Type 1, 2, and 3 components, respectively, of this study. Results from their IRM studies of 18 specimens lead them to suggest that the B component always resides in hematite, the A component is always held in magnetite, and that the steep component is a viscous overprint also in magnetite. As discussed above, the present study provides evidence in support of the first point, but differs somewhat from the latter two. Rather than existing solely in magnetite, the Type 3 component is found to be associated with both magnetite and hematite. Also, the Type 1 component, in some cases, appears to be a recent magnetization residing in goethite, while in other cases is most likely a viscous contribution to the NRM.

Significance of the Type 1 component

Similar to the Type 1 component, the steep component of Deutsch and Prasad (1987) occurs within 60% of the sampling suite used in their study. Although the authors do not list the mean vector direction of this component, it is steep and northerly directed and is a relatively "soft" component observed in the low-temperature range corresponding very well with the observed characteristics of the Type 1 component. Therefore, it is concluded that these directions represent a present day field overprint which either resides in a weathering product of these rocks (such as goethite or pyrrhotite) or is a viscous overprint, possibly in a magnetite phase.

Significance of the Type 2 component

While the Type 2 component was present in 39 of the 84 specimens in the present study, the B component was found in a total of only 14 specimens in the Deutsch and Prasad study. Also, the mean B vector direction is slightly more southeasterly and shallow than the Type 2 component. The in situ direction was determined to be $D = 168°$, $I = -9°$ with a corresponding antipole position located at 45°N, 138°E. The directional results

from the in situ Type 2 component are $D = 176°$, $I = -20°$, with an antipole located at 52° N, 128° E. The discrepancy between Type 2 and B component directions may arise from the difference in the number of observations included in the analyses of the mean or differences in the methodology used in calculating the specimen level directions. Deutsch and Prasad state that while half of the specimen directions used to calculate the mean B component were found by least square fitting of demagnetization trends, the second half was determined from the mean of vector end point groupings.

As pointed out in Deutsch and Prasad (1987), these directions are similar to those which would be anticipated for remanence acquired during the Kiaman superchron (Late Carboniferous–Early Permian; Irving and Strong, 1985). The mean Kiaman pole quoted by Irving and Strong (1985) is located at 44° N, 126° E. Assuming a Devonian age for the folding of the St. George Group (Schillereff and Williams, 1979; Cawood and Williams, 1988), if the Type 2 component (and B component) is of Kiaman age, it must represent a post-folding overprint of an earlier remanence. Therefore, for comparison with Permo-Carboniferous poles only the in situ directions should be considered.

As can be seen in Fig. 9, despite their differences the in situ north poles of the Type 2 and B components lie close to the mean Kiaman pole. Furthermore, the Type 2 pole is in excellent agreement with Permo-Carboniferous remagnetization components from the Mauch Chunk and Heldeberg formations of the U.S. Appalachians (Table 1 of Miller and Kent, 1988). For these formations, the remagnetization pole positions are located at 52° N, 130° E and 50° N, 129° E, respectively and while the Heldeberg pole was determined to be a post-folding overprint, that of the Mauch Chunk formation was shown to represent a synfolding remagnetization associated with Alleghanian folding.

Kiaman remagnetization of Paleozoic rocks within the Appalachian Orogen is a widely proposed explanation for certain anomalous pole positions (McCabe et al., 1983; Irving and Strong, 1985; Kent, 1985; Miller and Kent, 1986, 1988). Most authors invoke remagnetization mechanisms that include the migration of hot fluids, expelled from other tectonically deformed and mobilized areas of the orogen during the mid-late Paleozoic Alleghanian Orogeny (Oliver 1986). As pointed out by Piper (1987), Miller and Kent (1988), and many others, limestones and dolostones are highly susceptible to remagnetization associated with diagenetic processes. Miller and Kent (1988) have suggested that contrary to other types of lithologies (such as red beds), the remagnetization component within Paleozoic limestones of the Appalachians is usually the only one present after removal of the present day overprint. This observation is, in general, supported by the St. George data in that the Type 2 characteristic directions usually occur to the exclusion of the Type 3 component. In other words, with few exceptions, when the Type 2 component is observed, the Type 3 is not.

It has been proposed that the effects of the Alleghanian Orogeny are relatively minor in the northern Appalachians (i.e. Newfoundland) compared to those observed in the U.S. Appalachians (Williams and Hatcher, 1983). However, the geology of the St. George Group of Port-au-Port Peninsula provides evidence for Late Paleozoic uplift and circulation of fluids through subterranean conduits in the form of high-angle faults, collapse breccias and karstified surfaces of Carboniferous age which are common along the southern coast of the Peninsula (Schillereff and Williams, 1979; Knight and James, 1987) Strata containing these features are likely to bear the paleomagnetic signature of the ambient field at the time during which diagenetic changes may have occurred associated with the development of these features. This geologic evidence along with the similarity of the Type 2 component, the Mauch Chunk, Heldeberg and mean Kiaman poles make plausible the hypothesis that a portion of the remanence exhibited by the rocks of the lower St. George Group has been reset by a Late Paleozoic remagnetization. Although not conclusive, the fact that the Type 2 component is always found in association with hematite seems to support a chemical origin for this type of remanence (i.e. a CRM). Therefore, it is concluded here that the

Type 2 component represents a Kiaman overprint and that at the time of Type 2 remanence acquisition, this part of Port-au-Port peninsula was located at a paleolatitude of $10°N \pm 4°$. It is interesting to note that if this is correct, the processes leading to this remagnetization were pervasive, but not entirely complete. Although the Type 2 remanence is obviously an important component of the paleomagnetism of the St. George rocks, the fact that 40% of the specimens analyzed in this study apparently escaped its effects may attest to the non-uniform nature of such remagnetization phenomena.

Significance of the Type 3 component

In the Deutsch and Prasad study, 32 specimens exhibited the A component compared to 34 Type 3 specimens in this study. The mean directions for the A component were found to be $D = 150.9°$, $I = 19.6°$ (in situ) and $D = 148.5°$, $I = 33.6°$ (tilt corrected) which are very similar to the Type 3 component with regard to the paleo-inclinations, but are oriented in a more southeasterly direction than the Type 3 ($D = 161°$, $I = 16°$ in situ, $D = 163°$, $I = 34°$ with correction). The authors do not quote an in situ pole position for the A component, but the antipole position after structural correction is located at 17.5°N, 152°E. That of Type 3 is located in a slightly higher and more westerly position at 21°N, 138°E. This relatively minor discrepancy is obviously related to the apparent declination difference observed between the two determinations.

The map of the southern coast of Port-au-Port (Fig. 1) shows the declinations calculated from the Type 3 site level means (after structural correction; see also Table 1). It is clear that, within the level of resolution of the data, the declinations observed from each site are parallel, or sub-parallel to one another. There does appear to be a progression to more southwesterly orientations going from site NIB to site NSM (i.e. east to west), but this seems statistically insignificant given the level of accuracy of these determinations. Also, the site level directions from the two most westerly sites (NSH, NSM) are based upon one, and two specimen determinations, respectively. Furthermore, the Type 2 remanence is more frequently observed in the specimens of the more westerly sites, and it appears that the effects of the Kiaman remagnetization intensify from east to west. The determinations from the NSH and NSM sites are more shallow and slightly more south-southwesterly than those from the other sites and may actually represent the combined vector sum of both the Type 2 and Type 3 components. It seems likely that the sub-parallelism of these data either reflect: (1) the fact that the remanence is of pre-folding age and that this portion of the former continental shelf behaved in an integral, rather than fragmented manner during its deformation, (2) that the Type 3 directions represent a Taconic overprint, or (3) that the Type 3 component represents a post-folding (Devonian or younger) magnetization and that therefore, any local tectonic rotations between sampling sites are obscured.

The third possibility is rejected given the dissimilarity of the in situ Type 3 pole and the Devonian portion of the APW path for North America shown in Fig. 9. In fact, both the in situ and structurally corrected pole positions are in good agreement with the Early Ordovician part of the APW curve. This is due to the relatively minor amount of bedding correction applied. Unfortunately, the shallow dips and similar dip corrections of the bedding at the sampling sites renders inconclusive any test for discriminating between a Taconic vs. pre-Taconic age of remanence based solely on the structural correction applied. Acquisition of Type 3 remanence may have occurred at virtually any time subsequent to the Tremadocian formation age and prior to the Acadian Orogeny. It is unlikely that the magnetization post-dates the Acadian folding event, but there is approximately 100 million years between deposition of the St. George Group and its folding during which it could have been acquired. Although the deformation of the autochthonous stratigraphy on Port-au-Port is primarily attributed to Acadian tectonism (Schillereff and Williams, 1979; Cawood and Williams, 1988), the uncertain affects of the emplacement of the Humber Arm Allochthon upon this portion of the former continental margin further complicate the interpretation of the Type 3 component. Even

though the magnetic mineralogy associated with the Type 3 component includes magnetite (as well as hematite) as a major remanent phase, this does not provide compelling evidence for the age of remanence acquisition. This is because magnetite may be found to be either a primary (depositional) or authigenic mineralogical phase (McCabe et al., 1983).

Jackson and Van der Voo (1985) performed a paleomagnetic investigation of 149 specimens collected from a similarly aged carbonate unit, the Oneota dolomite of the Prairie du Chien Group of Minnesota, Iowa and Wisconsin. The Oneota dolomite is contained within the "inner dolomite facies" of Lower Paleozoic shelf carbonates of North America and their age is stratigraphically well defined as early Gasconadian (495–500 Ma; quoted in Jackson and Van der Voo, 1985). The sample suite yielded antipodal, normal and reversed directions and a positive breccia test, both generally viewed as evidence for an early acquired remanence. As such, the determination resulting in the Oneota pole (10.4° N, 166.4° E; pole O_1 in Fig. 9) is considered to be a reliable estimate of the direction of the Early Ordovician field and yields a paleolatitude of $1°S \pm 5°$ for the "inner dolomite" part of the Early Paleozoic shelf. The paleolatitude of Port-au-Port Peninsula has been recalculated using its present day geographical coordinates and the Oneota pole. The result derived is found to be $19°S \pm 5°$ which is exactly that found from the tilt corrected Type 3 directions ($19°S \pm 6°$). Therefore, based upon this comparison and the discussion above, it is reasonable to assume that the Type 3 component represents an early, prefolding Ordovician magnetization (not necessarily primary) of the lower St. George Group. It is possible that the Type 3 magnetization event may have been related to diagenetic processes accompanying exposure of the former carbonate shelf and the development of the Boat Harbour disconformity at the upper limit of the lower depositional megacycle of the St. George Group (Fig. 2; Knight and James, 1987). The inclusion of both magnetite and hematite in the magnetic mineralogy associated with the Type 3 component seems to support the speculation of a chemical origin for this magnetization. If correct, the Type 3 remanence would have been acquired toward the end of the Tremadocian.

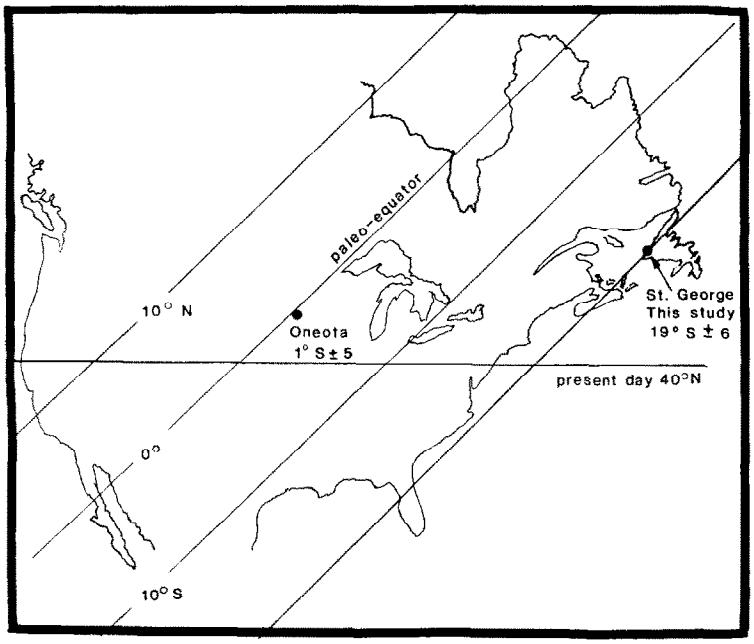


Fig. 10. Lines of paleolatitude for the Early Ordovician based on the results presented in this study and those of Jackson and Van der Voo (1985; Oneota) superimposed on a generalized map of present day North America.

However, to test this hypothesis, further paleomagnetic and petrographic work is required.

If the assumption that the St. George Type 3 and Oneota poles are characteristic of the Early Ordovician geomagnetic field is correct, the paleolatitudinal position of the Laurentian continent would be similar to that shown in Fig. 10. The paleo-equator is shown to pass through New Mexico, Nebraska, Minnesota, central Ontario, and northern Quebec. The southern continental margin (eastern; present day) would have located south of the equator and oriented E–W, subparallel to lines of latitude. This is in good agreement with the distribution of lower ordovician shallow-water carbonates within the Appalachian Orogen and fits well with the paleogeographic models for the Early Paleozoic of North America discussed earlier in regard to the tectonics and geology of the Northern Appalachians.

Conclusions

A paleomagnetic investigation of the Tremadocian portion of the Lower Ordovician St. George Group of Port-au-Port Peninsula, western Newfoundland indicates that these rocks are characterized by a three component remanence acquisition history. The most recently acquired component is northerly directed and steeply positive and corresponds well with the axial dipole field and the present day field at the sampling sites. This component, designated the Type 1 component is found to reside in the low T_{UB} range of the NRM and was generally removed by heating the specimens to about 300 °C. In some specimens, the Type 1 remanence appears to be associated with goethite, while in others it persists well beyond goethite T_{UB}'s and may best be explained as a viscous contribution to the NRM from a multi-domain magnetite phase.

The second, or Type 2 remanence component exists within a variable T_{UB} range, but was most often observed in the 450–600 °C range. Rock magnetic analyses indicated hematite to be the primary carrier of this Type 2 component. Specimens exhibiting Type 2 remanence are characterized by stable, linear high-temperature demagnetization trends corresponding to southerly oriented and negative (upward) pointing vector directions in situ. This magnetization direction is very similar to that which would be expected for remanence acquired during the Kiaman interval of Permo-Carboniferous times. It is concluded that the Type 2 component represents a Kiaman remagnetization of the lower St. George Group likely associated with Late Paleozoic orogenic events. As such, this part of the North American continent would have been situated slightly north of the paleo-equator at the approximate time of the Alleghanian Orogeny.

The effects of this remagnetization event appear to have been pervasive, but not complete within this portion of the St. George Group. Evidence for the incomplete nature of this event is the occurrence of the Type 3 remanence component. Type 3 directions were most often observed as linear decay segments in the 350–550 °C or 450–600 °C temperature range and were found to be contained in both magnetite and hematite mineral phases. The southeasterly and positive (downward) orientation of this component is seen to correspond well with reported Ordovician magnetization directions. The effects of the Taconic Orogeny on the St. George Group on Port-au-Port Peninsula and whether or not the Type 3 remanence acquisition is pre-Taconic is unclear. However, the mean vector direction is in good agreement with results from approximately coeval cratonal carbonate sequences elsewhere in North America shown to exhibit an early acquired remanence. The Type 3 remanence component is interpreted as a Early Ordovician (not necessarily primary), possibly pre-Taconic magnetization. This places the position of this portion of the former Laurentian margin of the Iapetus Ocean at a paleolatitude of about 20 °S prior to closure of this oceanic tract, and emplacement of the Humber Arm Allochthon. Finally, it appears that the Laurentian margin may have been approximately aligned with lines of paleolatitude in an east–west orientation during the Early Ordovician.

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